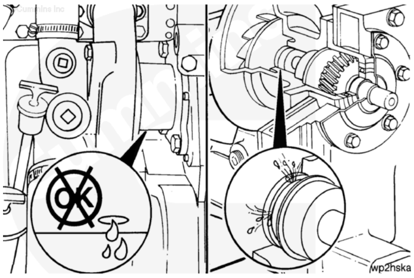
Trainings

Maintenance Check

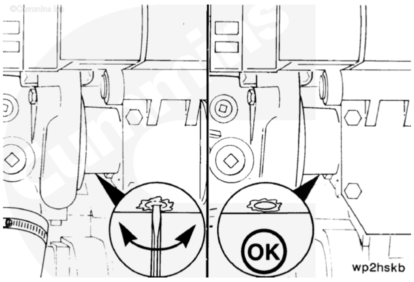
A streak or chemical buildup at the weep hole is **not** justification for water pump replacement.

If a steady flow or drip of coolant or oil is observed at the weep hole, replace the water pump with a new or rebuilt unit. Reference the Remove and Install sections of this procedure.



A small screwdriver or similair tool can be used to remove any debris.

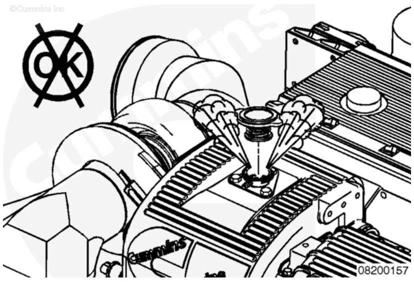
Make sure the weep hole is open.



Remove

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

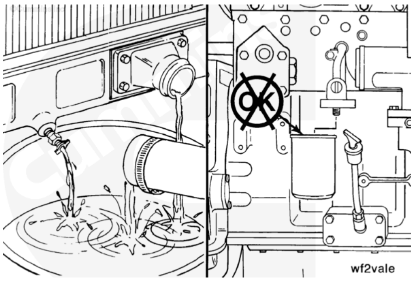


Remove the pressure cap when the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-om-mar.html)

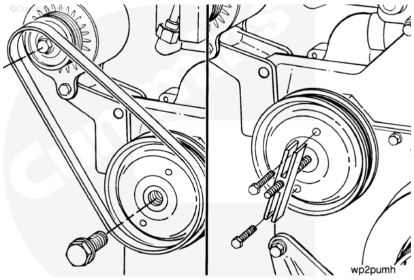


Note : Make sure the puller capscrews are threaded all the way through the puller before applying pressure to the puller screw.

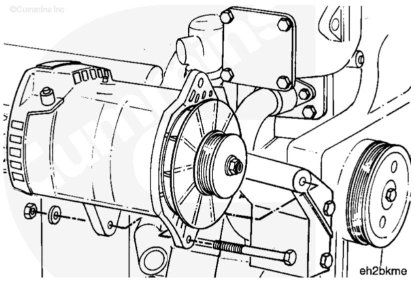
Remove the alternator drive belt. Refer to Procedure 013-005 in Section A. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-om-mar.html)

Remove the water pump pulley retaining capscrew.

Use the standard pulley puller, Part Number ST-647, or equivalent, and two 5/16 x 18 x 2 capscrews to remove the pulley.



Remove the alternator. Refer to a Cummins® Authorized Repair Location.

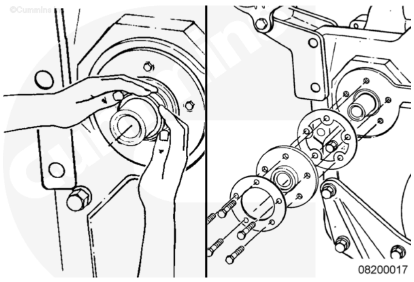


Note : Remove the dust seal as the seal carrier is removed, or use a heel bar, or similar tool, to pry the dust seal away from the seal case. Then remove the dust seal by hand.

Remove the dust seal.

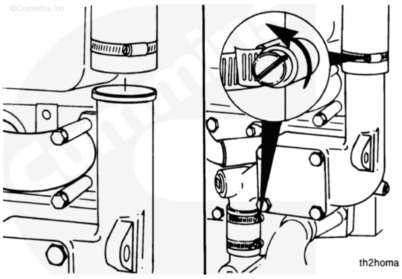
Remove the five water pump oil seal capscrews, clamping ring, oil seal, and gasket.

Discard the oil seal and dust seal.

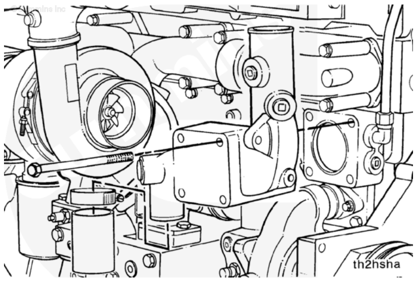


Loosen the coolant bypass hose clamps on both the upper and lower hoses.

Remove the upper coolant hose from the thermostat housing.

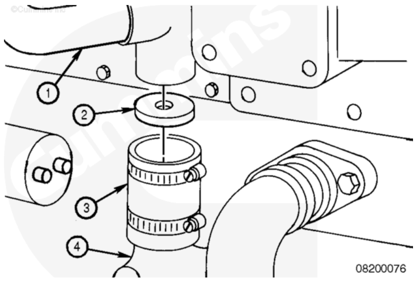


Remove the four thermostat housing mounting capscrews and the thermostat housing.



The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.



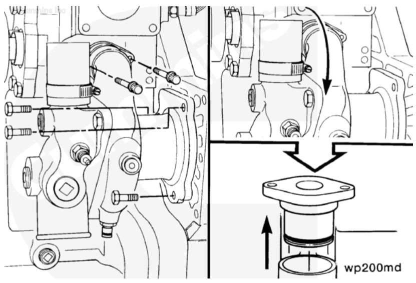
1. Torque converter coolant supply.
2. Torque converter disc (orifice).
3. Bypass hose.
4. Water pump.

Remove the two water pump water transfer connection capscrews.

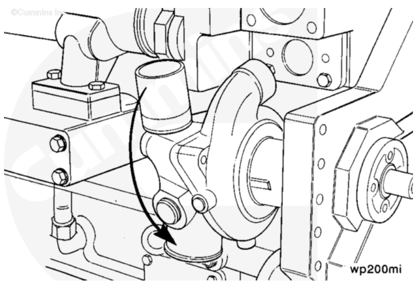
Remove the three water pump mounting capscrews.

Rotate the water pump outward so the water transfer connection can be removed from the water pump.

Remove the water transfer connection from the water pump.

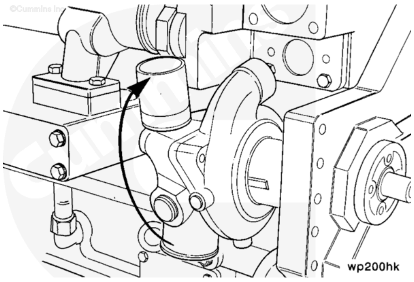


Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward, as it is being removed, to allow the pump to pass the thermostat housing support.



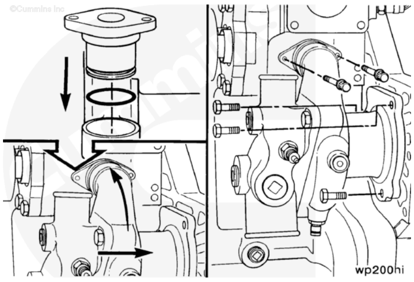
Install

Note : The water pump must be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.



Install a new o-ring on the water pump mounting flange.

Install the water pump.



Install a new o-ring on the water pump water transfer tube.

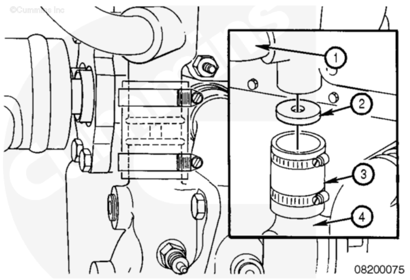
Install the connection into the water pump.

Twist the water pump inward, and install the three water pump mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]

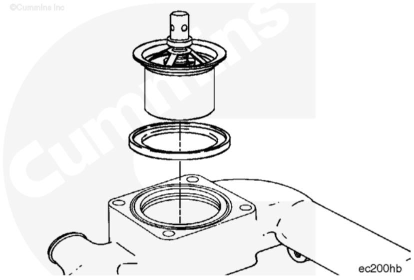
Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 n•m [221 in-lb]



If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter cooler coolant supply.
2. Torque converter cooler disc (orifice).
3. Bypass hose.
4. Water pump.



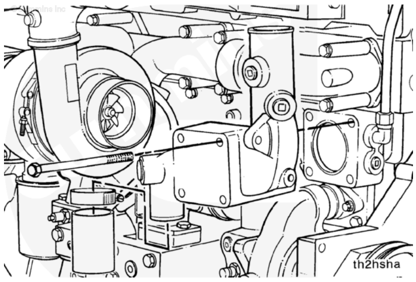
Install the thermostat in the housing.

Install a new seal in the groove on the thermostat housing mounting surface.

Install the hose on the thermostat housing bypass outlet.

Install the thermostat housing and four mounting capscrews.

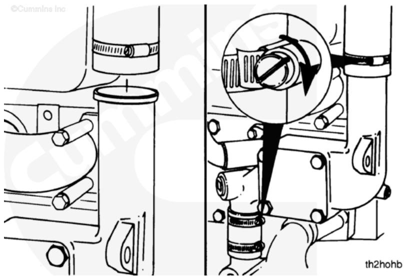
Torque Value: 54 n•m [40 ft-lb]



Equally space the bypass hose over the water pump connection and thermostat housing connection, and tighten the bypass hose clamps.

Torque Value: 3 n•m [27 in-lb]

Install the upper and lower coolant hoses. Refer to the OEM's specifications for the correct torque value.



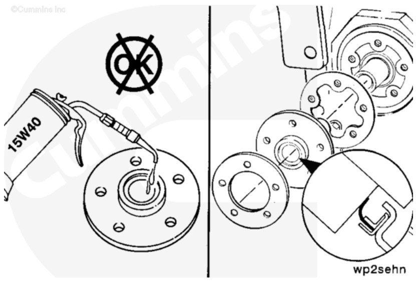
The oil seal **must** be installed with the lip of the seal and the shaft clean and dry. Do **not** lubricate. The yellow dust lip **must** be facing out.

Install the new gasket and oil seal. Use the installation sleeve provided with the new seal to install the seal.

The capscrew threads **must** be coated with thread sealant, Part Number 3823494 or equivalent, to prevent oil leakage.

Torque Value:

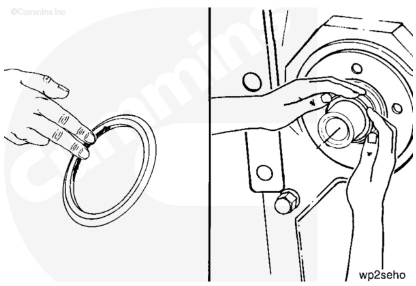
1. 7 n•m [62 in-lb]
2. 20 n•m [177 in-lb]



Place a light film of oil or antifreeze on the inside diameter of the dust seal.

Install the dust seal onto the shaft with the larger outside diameter facing the engine.

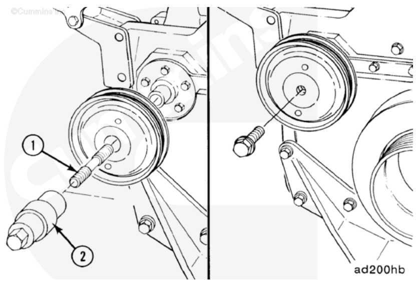
Push the dust seal back by hand on the shaft until the entire dust seal contacts the oil seal case.



Use pulley pusher adapter (1), Part Number 3377401, or equivalent, and pulley pusher (2), Part Number 3376326, or equivalent, to install the pulley.

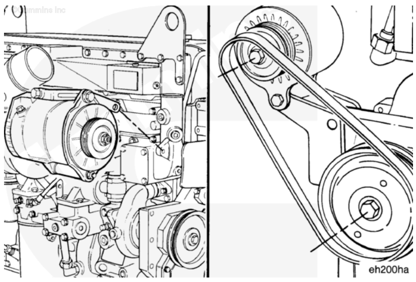
Install the capscrew in the shaft.

Torque Value: 75 n•m [55 ft-lb]



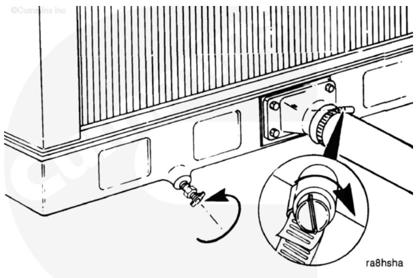
Install the alternator. Refer to a Cummins® Authorized Repair Location.

Install and adjust the alternator drive belt. Refer to Procedure 013-005 in Section A. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-om-mar.html)



Close the cooling system draincock, and install the lower coolant hose.

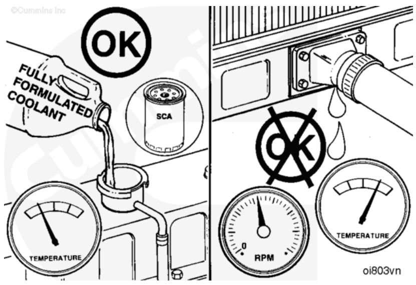
Tighten the hose clamps(s). Refer to the manufacturer's specifications for the correct torque value.



The correct concentration of coolant additives **must** be used in the cooling system. Refer to Procedure 018-004 in Section V (/qs3/pubsys2/xml/en/procedures/101/101-018-004.html).

Fill the cooling system.

Refer to Procedure 008-018 in Section 7 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-om-mar.html).



Last Modified: 03-Jan-2012